

Checkpoint 13: all-scale phase action and tensor normal-cone barriers

1. Candidate-proof status

There is still no complete dimension-free proof, and no KLS conclusion is claimed at this checkpoint. Four independent mechanisms have nevertheless been reduced to exact, dimension-free statements.

1. Tensoring a spectral extremizer creates an off-diagonal stress equal to $-\lambda f \otimes f$. If both signs of every moment-preserving potential variation were available, this would force f to be affine. The convex-potential normal cone is now shown to carry order-one, rather than perturbative, normalized action.
2. Gaussian observation of a balanced label has an exact monotone perimeter functional over all signal-to-noise scales. Its derivative is a nonnegative rigidity defect, but the associated unit action measure can migrate arbitrarily far along the logarithmic scale.
3. Bounded projected stochastic localization creates constant curvature outside finitely many weak directions. Killing successively more martingale coefficients does not close the argument: rotation and a signed third-moment drift survive, and explicit smooth log-concave tests rule out the most natural finite-dimensional Lyapunov corrections.
4. The singular zero-strain displacement branch is exactly a cyclic binary martingale dilation. The desired estimate becomes a dimension-free quadratic-variation bound for that dilation; local Hessian and entropy charges provably miss physical gaps.

All constants below are numerical and independent of the dimension.

2. Tensor-square spectral rigidity

Let μ be isotropic and let f be a normalized first eigenfunction,

$$-L_\mu f = \lambda f, \quad \int f d\mu = 0, \quad \int f^2 d\mu = 1.$$

On $\mu \otimes \mu$, the first eigenspace contains $F_1(x, y) = f(x)$ and $F_2(x, y) = f(y)$. Their mixed stress is exactly

$$\nabla F_1 \cdot \nabla F_2 - \lambda F_1 F_2 = -\lambda f(x)f(y). \quad (2.1)$$

At a two-sided interior minimizer, the full matrix first-variation equation says that every eigenspace stress is a polynomial of total degree at most two, modulo the barycenter and covariance constraints. Double centering then gives the dimension-free factorization lemma

$$\text{dist}_{L^2(\mu \otimes \mu)}(f \otimes f, \mathcal{P}_2) \leq \varepsilon \implies \text{dist}_{L^2(\mu)}(f, U)^2 \leq 1 - \sqrt{1 - \varepsilon^2}, \quad (2.2)$$

where U is the space of linear functions. Exact stationarity therefore makes f affine. On full space the eigenvalue equation then gives a standard Gaussian factor and $\lambda = 1$; on a bounded Neumann domain an affine first eigenfunction is impossible.

Convexity prevents promoting this argument directly. If $h = f \otimes f$, $P \in \mathcal{P}_2$, and

$$h - P = k_+ - k_- \quad (2.3)$$

with convex $k_\pm$, the new difference-of-convex audit proves

$$\frac{1}{\lambda} \|D^2 k_+ + D^2 k_-\|_{L^2(\mu \otimes \mu; \text{HS})} \geq \sqrt{2(1 - \lambda^2 |a|^4)} \geq \sqrt{2(1 - \lambda^2)}, \quad (2.4)$$

where $a = \mathbb{E}[Xf(X)]$. The proof isolates the mixed Hessian block $\nabla f(x) \nabla f(y)^T$; its best constant approximation is mm^T , with $m = \mathbb{E} \nabla f = \lambda a$. Thus along a hypothetical small-gap sequence the normalized convex buffer costs asymptotically $\sqrt{2}$, not $o(1)$.

There is also a dual action barrier. If a positive-semidefinite KKT measure N carries the normalized residual modulo $\mathcal{P}_2$, then

$$\int (D^2 k_+ + D^2 k_-) : dN \geq \|h - \Pi_{\mathcal{P}_2} h\|_2^2 = 1 - |a|^4. \quad (2.5)$$

The result holds for distributional Hessian measures and survives honest compact truncation. Therefore Bochner smallness of $D^2 f$ cannot make the normalized normal-cone defect disappear. Eliminating (2.5) would itself require new information of the strength of the missing full-matrix stationarity theorem.

3. The exact Target B reformulation

For $a \in \mathbb{R}^n$, define the directional negative Sobolev norm

$$H_\mu(a) = \sup_g \frac{(\int (a \cdot x) g d\mu)^2}{\int |\nabla g|^2 d\mu}. \quad (3.1)$$

It has the exact divergence formulation

$$H_\mu(a) = \min \left\{ \int |F|^2 d\mu : -\operatorname{div}(\mu F) = (a \cdot x)\mu \right\}. \quad (3.2)$$

The minimizing rows assemble the minimal Stein kernel τ_* , and

$$H_\mu(a) = a^T \mathbb{E}(\tau_* \tau_*^T) a. \quad (3.3)$$

If a normalized first eigenfunction satisfies $\operatorname{dist}(f, U) = \delta < 1$, its linear projection gives

$$1 - \delta^2 \leq \lambda H_\mu(a) \quad (3.4)$$

for a suitable unit direction a . Consequently the operator estimate

$$\mathbb{E}(\tau_* \tau_*^T) \preceq CI \quad (3.5)$$

would close the spectral stability target immediately. What is presently proved is only the trace estimate

$$\operatorname{tr} \mathbb{E}(\tau_* \tau_*^T) \leq Cn. \quad (3.6)$$

Moment-map estimates for $a^T \tau a$ do not bound $|\tau a|^2 = a^T \tau^2 a$; a positive rank-one matrix already separates the two quantities. Thus (3.5) is not a consequence of the known trace theorem and, because $H_\mu(a) \leq C_P(\mu)$, a growing genuine log-concave example would itself be KLS-relevant. The live replacement is to combine the near-linearity in (2.2) with the eigenfunction Bochner identity rather than assume (3.5).

4. Fixed-noise and all-scale Gaussian observation

Let E be balanced, put $U = 2\mathbf{1}_E(X) - 1$, and observe

$$Y_t = \sqrt{t} X + G, \quad G \sim N(0, I).$$

Write $p_t(y) = \mathbb{P}(E \mid Y_t = y)$, $F_t = \Phi^{-1}(p_t)$, and $I(s) = \varphi(\Phi^{-1}s)$. Every posterior law is t -strongly log-concave in the original variable. The sharp posterior covariance estimate gives

$$|\nabla F_t| \leq 1. \quad (4.1)$$

At unit noise the Bayes error $\operatorname{err}(E) = \mathbb{E} \min(p_1, 1 - p_1)$ obeys

$$\mu^+(E) \geq \sqrt{2/\pi} \operatorname{err}(E), \quad \operatorname{err}(E) \geq \frac{1}{4(C_P(\mu) + 2)}. \quad (4.2)$$

Thus a universal fixed-noise transition bound is quantitatively equivalent to KLS, not a weaker smoothing statement. The multiscale calculation yields the monotone perimeter functional

$$\mathcal{P}_E(t) = \sqrt{t} \mathbb{E} I(p_t), \quad \mathcal{P}'_E(t) = \frac{1}{2\sqrt{t}} \mathbb{E} [I(p_t)(1 - |\nabla F_t|^2)] \geq 0, \quad (4.3)$$

with endpoints

$$\mathcal{P}_E(0) = 0, \quad \mathcal{P}_E(\infty) = \mu^+(E). \quad (4.4)$$

For the posterior Bernoulli variance $B(t) = \mathbb{E}[p_t(1 - p_t)]$, one has

$$-B'(t) = t^{-1} \mathbb{E}[I(p_t)^2 |\nabla F_t|^2]. \quad (4.5)$$

Hence

$$d\omega_E(t) = 4t^{-1} \mathbb{E}[I(p_t)^2 |\nabla F_t|^2] dt \quad (4.6)$$

is a probability measure on logarithmic signal-to-noise scale, and

$$B(t_0) = \frac{1}{4} \omega_E([t_0, \infty)). \quad (4.7)$$

This is an exact unit-action decomposition of Boolean uncertainty. It does not locate the action. A uniform-interval model translates the whole profile by $-2 \log L$ under dilation; whitening deletes that elementary delay, but no proved identity pins a fixed positive fraction of ω_E to a universal scale for arbitrary isotropic log-concave measures. Such pinning would already imply (4.2).

The nuisance-cancelled low-signal expansion was also completed. With $Z = 2\mathbf{1}_E - 1$, let N_t be the signed output density relative to the standard Gaussian and L_t the unconditional output density. The correct chi-square information is

$$D(t) = \int \frac{N_t(y)^2}{L_t(y)} d\gamma(y) = \mathbb{E} [\mathbb{E}(Z | Y_t)^2]. \quad (4.8)$$

Writing $a = \mathbb{E}[ZX]$, $B = \mathbb{E}[ZXX^T]$, $R = \mathbb{E}[X^{\otimes 3}]$, and $K_3 = \mathbb{E}[ZX^{\otimes 3}] - 3\text{Sym}(a \otimes I)$, the exact expansion begins

$$\begin{aligned} D(t) &= t|a|^2 + t^2 \left(\frac{1}{2} \|B\|_{\text{HS}}^2 - |a|^2 \right) \\ &\quad + t^3 \left(|a|^2 - \|B\|_{\text{HS}}^2 + \frac{1}{6} \|K_3\|_{\text{HS}}^2 - \langle \text{Sym}(a \otimes B), R \rangle \right) + O(t^4). \end{aligned} \quad (4.9)$$

Formula (4.8) cancels independent asymmetric nuisance factors exactly. Separate conditional tensor estimates do not: an isotropic regular simplex has third-cumulant norm of order $\sqrt{n}$. More decisively, for a centered one-dimensional exponential law with its median label, $|\mathbb{E}[ZX^k]|$ grows factorially. The corresponding Hermite term has squared norm of order $k!$, and the $L^2(\gamma)$ chaos series has zero radius of convergence for every $t > 0$. Thus no fixed-degree analytic estimate can reach a numerical signal-to-noise ratio; the denominator in (4.8) would need a nonperturbative resummation.

5. Displacement midpoint: corrected covariance and determinant ledger

For the two normalized halves $\mu_+ = 2\mathbf{1}_E \mu$ and $\mu_- = 2\mathbf{1}_{E^c} \mu$, let T be the Brenier map and $D = T(X) - X$. If $K = \mathbb{E}DD^T$, then the midpoint law satisfies

$$\nu_{1/2} \leq 2\mu \quad (5.1)$$

and, after centering the half-measure displacement correctly,

$$\text{Cov}(\nu_{1/2}) = I - \frac{1}{4}K. \quad (5.2)$$

The previously tempting all-time affine covariance formula is false; the correct identity is

$$\text{Cov}(Z_t) = (1-t)C_+ + tC_- - t(1-t)K_c. \quad (5.3)$$

Using only domination, log-concavity, and one-dimensional moment bounds, the midpoint covariance has the universal spectral floor

$$\frac{1}{4800}I \preceq \text{Cov}(\nu_{1/2}) \preceq I. \quad (5.4)$$

Consequently

$$\frac{1}{4} \text{tr } K \leq -\log \det \text{Cov}(\nu_{1/2}) \leq 1200 \text{tr } K. \quad (5.5)$$

This is a useful absolute equivalence, but a determinant lower bound for every balanced transport remains KLS-equivalent. Moreover, covariance or log-determinant concavity along the geodesic is false. An explicit one-dimensional union-of-intervals example has zero entropy, AM–GM, and strain deficits for every t , while its midpoint variance is $(9\sqrt{5}-5)/16 < 1$. Translation gaps, not local strain, carry the cost.

6. Projected localization: what extra constraints do and do not buy

For a protected balanced label, let

$$b = \text{Cov}(\mathbf{1}_E, X), \quad u = b/|b|, \quad q = \text{Cov}((u \cdot Z)^2, X).$$

Killing both b and q in the driving matrix removes the direct martingale term in the protected directional variance, but the direction itself rotates. The exact remaining stochastic coefficient contains

$$\frac{2}{|b|}, CDP Au, \quad (6.1)$$

where $D = \mathbb{E}[(\mathbf{1}_E - p)ZZ^T]$ and $P = I - uu^T$. An anisotropic Gaussian with a balanced sign partition has $q = 0$ while (6.1) is nonzero. Thus a second scalar constraint does not freeze the weak line.

A rank-two kernel does give a deterministic curvature statement: after time t , the accumulated quadratic tilt is at least $t/3$ outside a subspace of dimension at most four. A static ball posterior nevertheless shows why this alone is insufficient. With curvature one outside a fixed two-plane and a nearly balanced sign label in that plane, the surviving variance and Poincare constant can both be of order $\sqrt{n}$.

The exact uniform-ball calculation also rules out a misleading dynamic counterexample. If

$$d\nu \propto e^{\langle c, x \rangle - \langle Bx, x \rangle / 2} \mathbf{1}_{|x| \leq \sqrt{n+2}} dx, \quad 0 \preceq B \preceq TI,$$

and v is an eigenvector of B , then

$$\text{Var}_\nu(v \cdot X) \leq \frac{6}{1-T}. \quad (6.2)$$

It follows from a log-concave transverse section and a one-dimensional divergence estimate. Therefore the apparent argmax-sign obstruction is critical only at $T = 1$; it cannot defeat a fixed subcritical projected flow.

Killing the full martingale coefficient of the protected variance leaves a signed third-moment drift. Even at $A = I$, it need not have a favorable sign. A smooth isotropic log-concave product of Gaussian-smoothed centered exponentials, with a diluted balanced label, makes this drift strictly positive while the drifts and quadratic variations of the natural

corrections $|b|^2$ and $b^T A^{-1}b$ are $O(\varepsilon^2)$. Consequently every bounded-derivative Lyapunov correction $r + \phi(|b|^2, b^T A^{-1}b)$, including fixed linear combinations, fails uniformly as $\varepsilon \downarrow 0$.

The function-specific version has now been closed to an exact ledger. Fix a normalized first eigenfunction f , let $\mathcal{E}_t = \int |\nabla f|^2 d\mu_t$, and choose $C_t b_t = 0$ for $b_t = \text{Cov}_{\mu_t}(f, X)$. Then

$$\mathbb{E} \text{Var}_{\mu_T} f = 1, \quad \mathbb{E} \mathcal{E}_T = \lambda. \quad (6.3)$$

For every relaxed hard driver with $\text{tr}(I - C_t^2) \leq 1$, the accumulated tilt has curvature $T/2$ outside one spectral direction v_T . If $R_T = \text{Var}_{\mu_T}(v_T \cdot X)$, the audited rank-one defect theorem gives

$$1 \leq \frac{192}{T} \lambda + 96 \mathbb{E}[R_T \mathcal{E}_T]. \quad (6.4)$$

Under the energy-biased path law $dQ = (\mathcal{E}_T/\lambda)d\mathbb{P}$, a small gap therefore forces $\mathbb{E}_{\mathbb{Q}} R_T \geq c/\lambda$. Conversely, the soft driver $C = I - \alpha P_b$, with $\delta = (1 - \alpha)^2$, has full curvature $B_T \geq \delta T I$ but obeys the exact mean/variance tradeoff

$$1 - \mathbb{E} m_T^2 \leq \frac{\lambda}{\delta T}. \quad (6.5)$$

Thus a universal bound on the energy-biased adaptive survivor in (6.4) would prove KLS, while enough soft curvature to bypass it necessarily loses the protected mean when λ is small. The calculation does not provide a weaker bootstrap.

7. The zero-strain branch as a cyclic binary dilation

For a uniform isotropic convex body K , the exact zero-gap midpoint data consist of a half-volume set R and a locally constant cyclic firmly nonexpansive map $b : R \rightarrow \mathbb{R}^n$ such that

$$K = (I - b)(R) \dot{\cup} (I + b)(R). \quad (7.1)$$

If Z is uniform on R and ε is a fair sign, then

$$U = Z + \varepsilon b(Z) \quad (7.2)$$

is uniform on K and is a martingale dilation. Iterating (7.2) until the chain first exits R gives

$$\mathbb{P}(\tau > m) = 2^{-m}, \quad \mathbb{E}|Z_\tau - Z_0|^2 = 2\mathbb{E}_R |b|^2. \quad (7.3)$$

Writing $S = \mathbb{E}_R[bb^T]$, isotropy gives the exact covariance split

$$\text{Cov}(R) = I - S, \quad \text{Cov}(K \setminus R) = I + S. \quad (7.4)$$

The desired absolute theorem is therefore

$$\text{tr } S \leq C. \quad (7.5)$$

Local singular-Hessian mass cannot prove (7.5). The realizable one-dimensional example

$$K = [0, 6], \quad E = [0, 1] \cup [2, 4] \quad (7.6)$$

uses translations $+1$ and $+2$, has zero source-wall functional, and after isotropic normalization satisfies

$$\mathbb{E}_R |b|^2 = \frac{1}{4}, \quad \text{Var}(R) = \frac{3}{4}. \quad (7.7)$$

The canonical firm extension sends part of the complementary set outside K , invalidating the natural ambient mixed-volume containment. A valid proof of (7.5) must charge the physical gaps between plateau components and use cyclic realizability, not merely pairwise firm nonexpansiveness.

8. Updated mechanism registry

The following routes are now blocked in their tested forms.

- A small difference-of-convex buffer derived only from the Bochner bound; its normalized curvature is bounded below by (2.4).
- Replacing the directional Stein-kernel operator estimate by the known trace estimate or by scalar moment-map control.
- Fixed-noise Gaussian smoothing without a new transition-mass theorem, and multiscale action without a scale-location principle. Fixed-degree low-signal chaos is also blocked by factorial mixed moments.
- Two protected scalar moments, or the full protected-variance martingale coefficient, without a new control of rotation and third moments.
- Static finite-defect curvature without control of the covariance in the adaptively selected defect space. The function-specific ledger (6.4) shows that its energy-biased form grows like $1/\lambda$ along every hypothetical small-gap sequence.
- Entropy, Cayley strain, source-wall Hessian mass, or a canonical firm extension as the sole charge in the zero-gap displacement branch.

The live incompatible routes are:

1. a function-specific projected localization which preserves the mean of a first eigenfunction and estimates only its terminal variance;
2. a Bochner/PDE rigidity theorem for a first eigenfunction already close to a linear function;
3. a nonperturbative Gaussian-information estimate using the exact denominator in (4.8); and
4. a global physical-gap or laminar-extrusion theorem for cyclic binary dilations.

No live route is counted as a proof until it supplies a dimension-free estimate for every log-concave measure and every admissible test function.

9. Audit state

The rank-one defect theorem from Checkpoint 12 remains clean-room verified. The identities in Sections 2–7 have been checked on the Gaussian, cube, isotropic simplex, isotropic crosspolytope, products of centered exponential laws, a radial exponential law, and the asymmetric interval construction where applicable. These checks expose no hidden dimension factor, but they also do not bridge any of the boxed missing global statements.

The next audit gate is deliberately binary. A candidate must either:

- turn the one-direction trace defect in function-specific localization into an averaged estimate with a numerical constant; or
- turn tensor near-linearity plus the Bochner identity into a numerical lower bound for λ , without invoking the unresolved directional H^{-1} estimate.

Only after one of those gates closes will the approximation, affine-support, all-test-function, and canonical-instantiation audits be restarted for a candidate full proof.